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JOB ID: 68bd7a34-94b9-4fd7-a077-26e0925fac73

RESULT PAGE: https://tygs.dsmz.de/user_results/show?guid=68bd7a34-94b9-4fd7-a077-26e0925fac73

Table 1: Phylogenies

Publication-ready versions of both the genome-scale GBDP tree and the 16S rRNA gene sequence tree can be customized and exported either in SVG (vector graphic) or PNG format from within the phylogeny viewers in your TYGS result page. For publications the **SVG format is recommended** because it is lossless, always keeps its high resolution and can also be easily converted to other popular formats such as PDF or EPS. Please follow the link provided above!

Table 2: Identification

The below list contains the result of the TYGS species identification routine.

Explanation of remarks that might occur in the below table:

remark [R1]: The TYGS type strain database is automatically updated on an almost daily basis. However, if a particular type strain genome is not available in the TYGS database, this can have several reasons which are detailed in the FAQ. You can request an extended 16S rRNA gene analysis via the 16S tree viewer found in your result page to detect **not yet genome-sequenced** type strains relevant for your study.

remark [R2]: > 70% dDDH value (formula d_4) and (almost) minimal dDDH values for gene-content formulae d_0 and d_6 indicate a potentially unreliable identification result and should thus be checked via the 16S rRNA gene sequence similarity. Such strong deviations can, in principle, be caused by sequence contamination.

remark [R3]: G+C content difference of > 1 % indicates a potentially unreliable identification result because within species G+C content varies no more than 1 %, if computed from genome sequences (PMID: 24505073).

Strain	Conclusion	Identification result	Remark
'Galaxy28-[Filter sequences by length on dataset 26]'	belongs to known species	<i>Staphylococcus aureus</i>	

Table 3: Pairwise comparisons of user genomes vs. type-strain genomes

The following table contains the pairwise dDDH values between your user genomes and the selected type-strain genomes. The dDDH values are provided along with their confidence intervals (C.I.) for the three different GBDP formulas:

- formula d_0 (a.k.a. GGDC formula 1): length of all HSPs divided by total genome length
- formula d_4 (a.k.a. GGDC formula 2): sum of all identities found in HSPs divided by overall HSP length
- formula d_6 (a.k.a. GGDC formula 3): sum of all identities found in HSPs divided by total genome length

Note: Formula d_4 is independent of genome length and is thus robust against the use of incomplete draft genomes. For other reasons for preferring formula d_4 , see the FAQ.

Query	Subject	d_0	C.I. d_0	d_4	C.I. d_4	d_6	C.I. d_6	Diff. G+C Percent
'Galaxy28-[Filter sequences by length on dataset 26].fasta'	<i>Staphylococcus aureus</i> DSM 20231	93.4	[90.7 - 95.3]	89.4	[87.0 - 91.4]	95.0	[93.0 - 96.4]	0.13
'Galaxy28-[Filter sequences by length on dataset 26].fasta'	<i>Staphylococcus aureus</i> subsp. <i>anaerobius</i> DSM 20714	86.2	[82.6 - 89.2]	80.1	[77.1 - 82.7]	88.1	[85.1 - 90.5]	0.12
'Galaxy28-[Filter sequences by length on dataset 26].fasta'	<i>Staphylococcus schweitzeri</i> FSA084	85.0	[81.3 - 88.1]	36.4	[34.0 - 38.9]	73.3	[69.8 - 76.5]	0.47
'Galaxy28-[Filter sequences by length on dataset 26].fasta'	<i>Allobacillus salarius</i> SKP4-8	12.7	[10.1 - 16.0]	34.9	[32.5 - 37.4]	13.2	[10.8 - 15.9]	6.01
'Galaxy28-[Filter sequences by length on dataset 26].fasta'	<i>Staphylococcus argenteus</i> MSHR1132	86.1	[82.5 - 89.1]	33.6	[31.1 - 36.1]	72.3	[68.8 - 75.5]	0.35
'Galaxy28-[Filter sequences by length on dataset 26].fasta'	<i>Staphylococcus simiae</i> CCUG 51256	23.4	[20.1 - 27.0]	23.3	[21.0 - 25.7]	22.3	[19.5 - 25.4]	0.69
'Galaxy28-[Filter sequences by length on dataset 26].fasta'	<i>Staphylococcus petrasii</i> subsp. <i>jettensis</i> CCUG 62657	16.6	[13.6 - 20.1]	22.8	[20.5 - 25.2]	16.6	[14.0 - 19.5]	0.54
'Galaxy28-[Filter sequences by length on dataset 26].fasta'	<i>Staphylococcus hominis</i> subsp. <i>novobiosepticus</i> CCUG 42399	17.2	[14.2 - 20.8]	22.6	[20.3 - 25.0]	17.1	[14.5 - 20.1]	1.32
'Galaxy28-[Filter sequences by length on dataset 26].fasta'	<i>Staphylococcus taiwanensis</i> NTUH-S172	16.6	[13.6 - 20.1]	22.2	[19.9 - 24.6]	16.6	[14.0 - 19.5]	0.39
'Galaxy28-[Filter sequences by length on dataset 26].fasta'	<i>Staphylococcus devriesei</i> CCUG 58238	15.9	[13.0 - 19.4]	21.7	[19.5 - 24.2]	16.0	[13.4 - 18.9]	0.56
'Galaxy28-[Filter sequences by length on dataset 26].fasta'	<i>Staphylococcus epidermidis</i> JCM 2414	16.9	[13.9 - 20.4]	21.5	[19.2 - 23.9]	16.8	[14.2 - 19.7]	0.59
'Galaxy28-[Filter sequences by length on dataset 26].fasta'	<i>Plantactinospira veratri</i> CGMCC 4.7143	16.9	[13.9 - 20.4]	21.0	[18.8 - 23.4]	16.8	[14.2 - 19.7]	1.38
'Galaxy28-[Filter sequences by length on dataset 26].fasta'	<i>Staphylococcus capitis</i> subsp. <i>urealyticus</i> DSM 6717	17.9	[14.8 - 21.4]	21.0	[18.8 - 23.5]	17.6	[15.0 - 20.6]	0.19

Table 4: Strains in your dataset

Joint dataset of automatically determined closest type strains (if this mode was chosen), manually selected type strains (if selected accordingly) and the provided user strains, if provided (marked in **yellow**).

Strain	Authority	Other deposits	Synonyms	Base pairs	Percent G+C	No. proteins	Goldstamp	Bioproject accession	Biosample accession	Assembly accession	IMG OID
<i>Staphylococcus taiwanensis</i> NTUH-S172	Lin et al. 2022	BCRC 81315; JCM 34726	<i>Staphylococcus taiwanensis</i>	2517 683	33.1	2312		PRJNA643904	SAMN15437803	GCA_020544305	
<i>Staphylococcus simiae</i> CCUG 51256	Pantůček et al. 2005	DSM 17636; NCTC 13838; CCM 7213; LMG 22723	<i>Staphylococcus simiae</i>	2597 811	32.0	2370	Gp0356989	PRJNA224116	SAMN05978013	GCF_002902085	
<i>Staphylococcus hominis</i> subsp. <i>novobiosepticus</i> CCUG 42399	Kloos et al. 1998	CIP 105719; ATCC 700236; DSM 15614; GTC 1228; R22	<i>Staphylococcus hominis</i> subsp. <i>novobiosepticus</i>	2422 222	31.4	2358	Gp0356985	PRJNA224116	SAMN05978006	GCF_002902465	
<i>Allobacillus salarius</i> SKP4-8	Yiamsombut et al. 2022	KCTC 33905; LMG 30016; TISTR 2499	<i>Allobacillus salarius</i>	2578 908	38.7	2546		PRJNA555754	SAMN12329004	GCA_007559425	
<i>Staphylococcus capitis</i> subsp. <i>urealyticus</i> DSM 6717	Bannerman and Kloos 1991	CIP 104192; ATCC 49326; CCUG 35142; MAW 8436	<i>Staphylococcus capitis</i> subsp. <i>urealyticus</i>	2467 992	32.9	2338	Gp0357001	PRJNA224116	SAMN05978025	GCF_002901925	
<i>Staphylococcus devriesei</i> CCUG 58238	Supré et al. 2010	DSM 25293; NCTC 13828; KS-SP 60; LMG 25332	<i>Staphylococcus devriesei</i>	2379 863	33.3	2328	Gp0357004	PRJNA224116	SAMN05977994	GCF_002902625	

Strain	Authority	Other deposits	Synonyms	Base pairs	Percent G+C	No. proteins	Goldstamp	Bioproject accession	Biosample accession	Assembly accession	IMG OID
<i>Staphylococcus schweitzeri</i> FSA084	Tong et al. 2015	CCUG 70918; DSM 28300; NCTC 13712; SSI 89.004	<i>Staphylococcus schweitzeri</i>	2748 403	32.2	2523	Gp0102096	PRJEB6391	SAMEA2008002	GCA_000751255	
<i>Staphylococcus petrasii</i> subsp. <i>jettensis</i> CCUG 62657	(De Bel et al. 2013) De Bel et al. 2014	DSM 26618; NCTC 13830; CCM 8494; LMG 26879; SEQ110	<i>Staphylococcus jettensis</i> ; <i>Staphylococcus petrasii</i> subsp. <i>jettensis</i>	2709 761	33.2	2509	Gp0357019	PRJNA224116	SAMN05978011	GCF_002902105	
<i>Staphylococcus aureus</i> subsp. <i>anaerobius</i> DSM 20714	De La Fuente et al. 1985	CIP 103780; ATCC 35844; CCUG 37246; MVF-7	<i>Staphylococcus aureus</i> subsp. <i>anaerobius</i>	2575 736	32.8	2233	Gp0356999	PRJNA339206	SAMN05978010	GCA_002902425	
<i>Staphylococcus aureus</i> DSM 20231	Rosenbach 1884 emend. Madhaiyan et al. 2020	533 R4; CIP 65.8; NCCB 72047; ATCC 12600; ATCC 12600-U; CCUG 1800; JCM 20624; NBIMCC 746; NBRC 100910; NCTC 8532; CCM 885; HAMBI 66; NCAIM B.01065	<i>Staphylococcus aureus</i> ; <i>Staphylococcus aureus</i> subsp. <i>aureus</i>	2782 562	32.8	2734	Gp0120765	PRJNA283471	SAMN03651231	GCA_001027105	
<i>Staphylococcus argenteus</i> MSHR1132	Tong et al. 2015	CCUG 70917; DSM 28299; NCTC 13711; SSI 89.005	<i>Staphylococcus argenteus</i>	2787 636	32.4	2531	Gp0014226	PRJEA62885	SAMEA2272597	GCA_000236925	
<i>Plantactinospora veratri</i> CGMCC 4.7143	Xing et al. 2015	DSM 46718; NEAU FHS4	<i>Plantactinospora veratri</i>	2172 717	31.3	2115		PRJDB10509	SAMD00245180	GCA_014645875	

Strain	Authority	Other deposits	Synonyms	Base pairs	Percent G+C	No. proteins	Goldstamp	Bioproject accession	Biosample accession	Assembly accession	IMG OID
<i>Staphylococcus epidermidis</i> JCM 2414	(Winslow and Winslow 1908) Evans 1916	CIP 81.55; ATCC 14990; CCUG 18000 A; CCUG 18000; CCUG 39508; DSM 20044; NBIMCC 1093; NBRC 100911; NCTC 11047; CCM 2124; LMG 10474; NCAIM B.01066	<i>Albococcus epidermidis</i> ; <i>Staphylococcus epidermidis</i>	2302 683	32.1	2171	Gp0093194	PRJDB677	SAMD00000465	GCA_001310935	
Galaxy28-[Filter sequences by length on dataset 26].fasta				2889 728	32.7	2723					

Methods, Results and References

The genome sequence data were uploaded to the Type (Strain) Genome Server (TYGS), a free bioinformatics platform available under <https://tygs.dsmz.de>, for a whole genome-based taxonomic analysis [1]. The analysis also made use of recently introduced methodological updates and features [2,3]. Information on nomenclature, synonymy and associated taxonomic literature was provided by TYGS's sister database, the List of Prokaryotic names with Standing in Nomenclature (LPSN, available at <https://lpsn.dsmz.de>) [2,3]. The results were provided by the TYGS on 2026-05-20. The TYGS analysis was subdivided into the following steps:

Determination of closely related type strains

Determination of closest type strain genomes was done in two complementary ways: First, all user genomes were compared against all type strain genomes available in the TYGS database via the MASH algorithm, a fast approximation of intergenomic relatedness [4], and, the ten type strains with the smallest MASH distances chosen per user genome. Second, an additional set of ten closely related type strains was determined via the 16S rDNA gene sequences. These were extracted from the user genomes using RNAmmer [5] and each sequence was subsequently BLASTed [6] against the 16S rDNA gene sequence of each of the currently 24411 type strains available in the TYGS database. This was used as a proxy to find the best 50 matching type strains (according to the bitscore) for each user genome and to subsequently calculate precise distances using the Genome BLAST Distance Phylogeny approach (GBDP) under the algorithm 'coverage' and distance formula d_5 [7]. These distances were finally used to determine the 10 closest type strain genomes for each of the user genomes.

Pairwise comparison of genome sequences

For the phylogenomic inference, all pairwise comparisons among the set of genomes were conducted using GBDP and accurate intergenomic distances inferred under the algorithm 'trimming' and distance formula d_5 [7]. 100 distance replicates were calculated each. Digital DDH values and confidence intervals were calculated using the recommended settings of the GGDC 4.0 [2,7].

Phylogenetic inference

The resulting intergenomic distances were used to infer a balanced minimum evolution tree with branch support via FASTME 2.1.6.1 including SPR postprocessing [8]. Branch support was inferred from 100 pseudo-bootstrap replicates each. The trees were rooted at the midpoint [9] and visualized with PhyD3 [10].

Type-based species and subspecies clustering

The type-based species clustering using a 70% dDDH radius around each of the 13 type strains was done as previously described [1]. The resulting groups are shown in Table 1 and 4. Subspecies clustering was done using a 79% dDDH threshold as previously introduced [11].

Results

Type-based species and subspecies clustering

The resulting species and subspecies clusters are listed in Table 4, whereas the taxonomic identification of the query strains is found in Table 1. Briefly, the clustering yielded 11 species clusters and the provided query strains were assigned to 1 of these. Moreover, user strains were located in 1 of 11 subspecies clusters.

Figure caption SSU tree

Figure 1. Tree inferred with FastME 2.1.6.1 [8] from GBDP distances calculated from 16S rDNA gene sequences. The branch lengths are scaled in terms of GBDP distance formula d_5 . The numbers above branches are GBDP pseudo-bootstrap support values > 60 % from 100 replications, with an average branch support of 70.8 %. The tree was rooted at the midpoint [9].

Figure caption genome tree

Figure 2. Tree inferred with FastME 2.1.6.1 [8] from GBDP distances calculated from genome sequences. The branch lengths are scaled in terms of GBDP distance formula d_5 . The numbers above branches are GBDP pseudo-bootstrap support values > 60 % from 100 replications, with an average branch support of 56.4 %. The tree was rooted at the midpoint [9].

References

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